

Satyaki Sikdar

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Research Interests

Network Science · Science of Science · Social and Information Networks · Machine Learning

Education

University of Notre Dame **Notre Dame, IN**
Ph.D. in Computer Science & Engineering *2017 – 2022*

Adviser: Tim Weninger

Thesis: Scalable and Interpretable Graph Modeling with Graph Grammars

Committee: David Chiang, Peter Kogge, and Danai Koutra

Heritage Institute of Technology **Kolkata, India**
B.Tech. in Computer Science & Engineering *2013 – 2017*

Advisers: Partha Basuchowdhuri and Subhashis Majumder

Thesis: Learning Models for Influence Maximization

Work Experience

Loyola University Chicago **Chicago, IL**
Assistant Professor (tenure-track), Computer Science Department *Aug 2024 –*

Indiana University **Bloomington, IN**
Postdoctoral Fellow with Santo Fortunato, Luddy School of Informatics *Jan 2022 – Jul 2024*

University of Notre Dame **Notre Dame, IN**
Instructor of Record *Fall 2021, Spring 2021*
Graduate Research Assistant with Tim Weninger *May 2018 – Dec 2021*
Graduate Teaching Assistant *Aug 2017 – May 2018*

Snap Inc. **Remote**
Research Intern, Computational Social Science Team *May – Aug 2020*

Heritage Institute of Technology **Kolkata, India**
Undergraduate Researcher with Partha Basuchowdhuri and Subhashis Majumder *Apr 2015 – Jul 2017*

Honors and Awards

Teaching Award: Outstanding Graduate Student Instructor Award 2022

Student Award: Best Student Award for Academic Excellence – Class of 2017

Travel Awards: ACM WSDM 2022 · ACM IKDD Conference on Data Science 2016

Courses Taught

(† online)

Loyola University Chicago	Chicago, IL
COMP 306/406 Data Mining	Spring 2025
Class size: 34. Overall Course Effectiveness median score: 5 / 5.	
COMP 388/488 Network Science†	Spring 2025
Class size: 15. Overall Course Effectiveness median score: 4.85 / 5.	
COMP 306/406 Data Mining†	Fall 2024
Class size: 29. Overall Course Effectiveness median score: 4.5 / 5.	
COMP 141 Introduction to Computing Tools and Techniques†	Fall 2024
Class size: 28. Overall Course Effectiveness median score: 4 / 5.	
Network Science Society	
Co-instructor for A Short First Course in Network Science†	Summer 2024
A two-week online course co-taught with Santo Fortunato	
University of Notre Dame	Notre Dame, IN
CSE 30151 Theory of Computing	Fall 2021
Class size: 91. Composite median score: 4.7 / 5.	
CSE 20110 Discrete Mathematics	Spring 2021
Class size: 50. Composite median score: 3.7 / 5.	

Publications

(* equal contribution AR acceptance rate IF impact factor)

Journal Articles.....

- [J4] **S. Sikdar***, S. Venturini*, M. L. Charpignon, S. Kumar, F. Rinaldi, F. Tudisco, S. Fortunato, and M. S. Majumder, "What we should learn from pandemic publishing," *Nature Human Behaviour (comment)*, 2024.
- [J3] **S. Sikdar***, S. Venturini*, F. Rinaldi, F. Tudisco, and S. Fortunato, "Collaboration and topic switches in science," *Scientific Reports*, 2024, [IF:4.6].
- [J2] **S. Sikdar**, D. G. Cedre, T. W. Ford, and T. Weninger, "The Infinity Mirror Test for Graph Models," *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 2022, [IF:8.9].
- [J1] P. Basuchowdhuri, **S. Sikdar**, V. Nagarajan, K. Mishra, S. Gupta, and S. Majumder, "Fast Detection of Community Structures using Graph Traversal in Social Networks," *Knowledge and Information Systems (KAIS)*, 2018, [IF:2.7].

Conference Papers.....

- [C6] **S. Sikdar**, N. Shah, and T. Weninger, "Attributed Graph Modeling with Vertex Replacement Grammars," *ACM International Conference on Web Search and Data Mining (WSDM)*, 2022, [AR:18%].
- [C5] J. I. Hibshman, D. G. Cedre, **S. Sikdar**, and T. Weninger, "Joint Subgraph-to-Subgraph Transitions: Generalizing Triadic Closure for Powerful and Interpretable Graph Modeling," *ACM International Conference on Web Search and Data Mining (WSDM)*, 2021, [AR:18%].
- [C4] J. I. Hibshman, **S. Sikdar**, and T. Weninger, "Towards Interpretable Graph Modeling with Vertex Replacement Grammars," *IEEE International Conference on Big Data (BigData)*, 2019, [AR:19%].
- [C3] **S. Sikdar**, J. Hibshman, and T. Weninger, "Modeling Graphs with Vertex Replacement Grammars," *IEEE International Conference on Data Mining (ICDM)*, 2019, [AR:9%].
- [C2] C. Pennycuff, **S. Sikdar**, C. Vajiac, D. Chiang, and T. Weninger, "Synchronous Hyperedge Replacement Graph Grammars," *International Conference on Graph Transformation (ICGT)*, 2018.
- [C1] P. Basuchowdhuri, **S. Sikdar**, S. Shreshtha, and S. Majumder, "Detecting Community Structures in Social Networks by Graph Sparsification," *ACM IKDD Conference on Data Science (CODS)*, 2016, [AR:5%].

Peer-reviewed Abstracts

- [A7] S. Venturini, **S. Sikdar**, M. Mazzarello, F. Rinaldi, F. Tudisco, P. Santo, S. Fortunato, and C. Ratti, *"The Effect of Colocation on Paper Coauthorships during COVID-19,"* International Conference on Network Science (NetSci), Contributed Talk, 2024.
- [A6] S. Venturini, **S. Sikdar**, M. Mazzarello, F. Rinaldi, F. Tudisco, P. Santo, S. Fortunato, and C. Ratti, *"The Effect of Colocation on Paper Coauthorships during COVID-19,"* International Conference on Science of Science and Innovation (ICSSI), Poster, 2024.
- [A5] **S. Sikdar***, S. Venturini*, M. L. Charpignon, S. Kumar, F. Rinaldi, F. Tudisco, S. Fortunato, and M. S. Majumder, *"The COVID-19 Research Outbreak: How the Pandemic Culminated in a Surge of New Researchers,"* International Conference on Computational Social Science (IC²S²), Contributed Talk, 2023, [AR:31%].
- [A4] **S. Sikdar***, S. Venturini*, M. L. Charpignon, S. Kumar, F. Rinaldi, F. Tudisco, S. Fortunato, and M. S. Majumder, *"The COVID-19 Research Outbreak: How the Pandemic Culminated in a Surge of New Researchers,"* International Conference on Science of Science and Innovation (ICSSI), Contributed Talk, 2023.
- [A3] **S. Sikdar***, S. Venturini*, F. Rinaldi, F. Tudisco, and S. Fortunato, *"Collaboration and Topic Switches in Science,"* International Conference on Computational Social Science (IC²S²), Plenary Talk, 2023, [AR:1.75%].
- [A2] **S. Sikdar***, S. Venturini*, F. Rinaldi, F. Tudisco, and S. Fortunato, *"Collaboration and Topic Switches in Science,"* International Conference on Science of Science and Innovation (ICSSI), Contributed Talk, 2023.
- [A1] **S. Sikdar***, S. Venturini*, F. Rinaldi, F. Tudisco, and S. Fortunato, *"Collaboration and Topic Switches in Science,"* International Conference on Network Science (NetSci), Contributed Talk, 2023.

Contributed Chapters

- [R2] **S. Sikdar**, *"NetworkX Graph Library," A Survey of Graph Processing Paradigms*, P. M. Kogge, Ed., University of Notre Dame, 2019. DOI: 10.7274/r0-z6dc-9c71.
- [R1] **S. Sikdar**, *"Spectral Community Detection," A Survey of Graph Kernels*, P. M. Kogge, Ed., University of Notre Dame, 2019. DOI: 10.7274/r0-e7wb-da60.

Under Preparation

- [P1] **S. Sikdar***, A. Ghavasieh*, M. De Domenico, and S. Fortunato, *"Persistent Energy Localization Determines the Mesoscale Functional Organization of Complex Networks,"* Forthcoming, 2024.

Presentations

(* virtual talk)

Invited Talks

12 th Greater Chicago Area Systems Research Workshop, Chicago, IL	May 2025
Vidyasagar College, Kolkata, India	Jan 2025
Loyola University Chicago, Chicago, IL	Mar 2024
University of Minnesota, Duluth, MN	Feb 2024
University of Dayton, Dayton, OH	Feb 2024
DePaul University, Chicago, IL	Feb 2024
Heritage Institute of Technology, Kolkata, India	Sep 2022
Indian Association for the Cultivation of Science, Kolkata, India	Sep 2022

University of Louisville, Louisville, KY	Mar 2022
Lawrence Livermore National Laboratory, Livermore, CA*	Apr 2021
University of Illinois, Urbana, IL*	Apr 2021
Missouri State University, Springfield, MO*	Feb 2021
South Dakota State University, Brookings, SD*	Dec 2020
Indian Association for the Cultivation of Science, Kolkata, India	Oct 2019
Vidyasagar College, Kolkata, India	Jan 2016

Posters

<i>Cyberbullying Networks: Analyzing User Roles and Interactions on Instagram.</i> GCASR 2025	May 2025
<i>The Effect of Colocation on Paper Coauthorships during COVID-19.</i> ICSSI 2024	July 2024
<i>Consistency Pays off in Science.</i> ICSSI 2023	June 2023
<i>The Infinity Mirror Test for Graph Generators.</i> SIAM NS 2020 	July 2020
<i>Modeling Graphs with Vertex Replacement Grammars.</i> SIAM NS 2019, NetSci 2019 	May 2019
<i>Synchronous Hyperedge Replacement Graph Grammars.</i> MSLD 2018 	May 2018

Other Talks

<i>Introduction to Support Vector Machines</i> 	Apr 2017
<i>Density Based Spatial Clustering of Applications with Noise</i> 	Feb 2017
<i>Introduction to Decision Trees</i> 	Nov 2016
<i>A Friendly Introduction to Random Networks</i> 	Feb 2016

Service

Organizing Committee Member. Networks in Science of Science Workshop @ NetSci 2025

Program Committee Member.

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|---------------------------------|--------------------------------|----------------------|
| - IEEE ICDM 2025 | - WSDM 2025, 2024 | - BigData 2024, 2019 |
| - PAKDD 2025, 2024 | - ICWSM 2020, 2022, 2023, 2024 | - WISE 2019 |
| - The Web Conference 2025, 2024 | - CIKM 2021, 2022, 2023, 2024 | |
| - SDM 2022, 2023, 2024, 2025 | - ISSI 2023 | |

Journal Reviewer.

- | | |
|--|---|
| - Quantitative Science Studies | - PLOS ONE |
| - Scientometrics | - ACM Transactions on Data Science (TDS) |
| - EPJ Data Science | - Data Mining and Knowledge Discovery (DMKD) |
| - IEEE Transactions on Knowledge and Data Engineering (TKDE) | - International Journal Cooperative Information Systems (IJCIS) |
| - Journal of Combinatorial Optimization | - Journal of Logic & Computation |
| - Physical Review E | |
| - Physical Review Materials | |

External Reviewer (multiple times). IEEE ICDM, CIKM, The Web Conference, WSDM, KDD, AAAI

International Committee Leader. Postdoc Association Indiana University Bloomington 2022 – 2024

Volunteer. The Scoop from Grad Students Dec 2019

<i>Panelist.</i> Preparing a Grad School Application	Dec 2019
<i>Member.</i> Notre Dame CSE Graduate Student Board	2019 – 2020
<i>Volunteer Judge.</i> Northern Indiana Regional Science & Engineering Fair	2019, 2020
<i>Chair, Vice Chair, Secretary.</i> Heritage Institute of Technology ACM Student Chapter	2014 – 2017

Students Mentored & Supervised

(@ indicates co-authorship in papers)

Thesis Advisor

<i>Izzy Moore.</i> Ph.D. student, LUC	May 2025 –
<i>Juan Rosendo.</i> M.S. student, LUC (co-supervised with Yasin Silva)	Aug 2025 –

Research Mentor

<i>Taylor Hales.</i> M.S. student, LUC	May 2025 –
<i>Chloe Kilroy.</i> B.S. student, LUC	Jan 2025 –
<i>Tyler Rosario.</i> M.S. student, LUC	Aug 2024 – May 2025
<i>Manuel Sandoval Madrigal.</i> Ph.D. student, LUC	Aug 2024 –
<i>Andrew Claros.</i> Ph.D. student, IU	Aug 2023 –
<i>Marie-Laure Charpignon</i> [@] . Ph.D. student, MIT	Aug 2022 –
<i>Sagar Kumar</i> [@] . Ph.D. student, Northeastern	Aug 2022 –
<i>Arsham Ghavasieh</i> [@] . Ph.D. student, University of Trento, Italy (now Postdoc at IU)	Aug 2022 –
<i>Şirag Erkol</i> [@] . Ph.D. student, IU (now Postdoc at Northwestern)	Feb – Aug 2022
<i>Daniel Kaiser.</i> Ph.D. student, IU	Feb – Aug 2022
<i>Sara Venturini</i> [@] . Ph.D. student, University of Padova, Italy (now Postdoc at Northeastern)	Jan 2022 –
<i>Manita Pote.</i> Ph.D. student, IU	Jan – May 2022
<i>Daniel Gonzalez Cedre</i> [@] . Ph.D. student, ND	Dec 2019 – Aug 2020
<i>Catalina Vajiac</i> [@] . B.S. student, ND (now Ph.D. student at CMU)	Jan – Jul 2019
<i>Justus Hibshman</i> [@] . Ph.D. student, ND	Sep – Dec 2018

Computer Skills

Advanced: Python, NetworkX, Matplotlib, Seaborn, Jupyter, $\text{\LaTeX}$
Intermediate: C++, Boost Graph Library, Pandas, SQL, shell scripting

Relevant Coursework

Scalable Graph Algorithms · Web Science & Information Retrieval · Exotic Computing · Social Networks Complexity & Algorithms · Pedagogy and Practice in the College Classroom

References

Tim Weninger	574-631-6770
Frank M. Freimann Collegiate Associate Professor of Engineering Department of Computer Science & Engineering University of Notre Dame	tweninger@nd.edu
Santo Fortunato	812-856-5754
Professor APS Fellow, Network Science Society Fellow Luddy School of Informatics, Computing, and Engineering Indiana University Bloomington	santo@iu.edu
Peter M. Kogge	574-631-6763
Ted H. McCourtney Professor of Computer Science & Engineering IBM Fellow, IEEE Fellow, AAAS Fellow University of Notre Dame	kogge@nd.edu
Maimuna S. Majumder	617-355-6000
Assistant Professor Peter Szolovits Distinguished Scholar Computational Health Informatics Program Harvard Medical School & Boston Children's Hospital	maimuma.majumder@childrens.harvard.edu